

Training by the Research Computing Group



SIMON FRASER
UNIVERSITY



**Digital Research
Alliance** of Canada

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<https://training.westdri.ca/contact>

Training team



MARIE-HÉLÈNE BURLE
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- Background in evolutionary & behavioural ecology (SFU)
- 18 years training and support experience (university courses, Research / Student Learning Commons, WestGrid, RCG)
- Passionate open-source advocate, certified Software Carpentry instructor



ALEX RAZOUMOV
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- Background in computational astrophysics (UBC)
- Postdocs in Urbana-Champaign, San Diego, Oak Ridge, & Halifax
- 30+ years training and support experience (university courses, SHARCNET, WestGrid, RCG)
- National Visualization Team lead since 2014, certified Software Carpentry instructor

Both can be reached at training@westdri.ca

Research computing training at SFU and beyond

- Our funding:



supports training researchers in their actual computational workflows, to be run on our systems (max science, max hardware utilization)



does not support undergraduate academic programs (teaching university courses)

- In practice, we often see graduate students lacking basic computational skills, highlighting the need for earlier training and closer collaboration with departments

- SFU is one of the Alliance's national host sites
⇒ our training and support are available to all academic researchers across Canada

- anyone with a valid Canadian university email address can register for our courses
- webinars and online materials open to anyone
- we answer national support tickets and contribute to the national docs <https://docs.alliancecan.ca>



Topics: full-day courses

Details/abstracts at <https://training.westdri.ca/courses>

● Remote computing basics

- Bash command line
- Introduction to HPC
- + Specific remote computing tools

● Programming basics

- Python • R • Julia
- Scientific Python
- Webscraping or GIS in Python or R
- Selected Python libraries: Polars, PyTables, ...

● Parallel coding (next slide)

- Julia • Chapel • Python • R
- GPU computing with Chapel, Python, Julia

● Deep learning

- PyTorch • JAX+Flax

● Virtualization

- Intro to Apptainer containers
- Intro to cloud VMs

● Scientific visualization (15+ workshops)

- ParaView • VisIt • 3D vis for the humanities
- Python/R plotting libraries
- In-situ vis, remote and large-scale parallel rendering, scripting, animation, interactive web visualization, other advanced sci-vis topics

● RDM topics

- Version control with Git
- Distributed datasets: git annex, or DataLad, or DVC
- Managing files and formats: scientific file formats, parallel I/O, container overlays, etc.

● Publishing

- R Markdown and Quarto • Interactive dashboards

Parallel coding: our bread and butter

- 10-15 years ago we used to teach MPI (message-passing interface) and OpenMP in C/C++/Fortran (most common HPC languages)
- Today's "Intro to HPC" covers MPI and OpenMP when discussing parallel frameworks
 - but we don't teach actual coding in this course
 - focus on basic cluster tools, software env, running codes (serial, shared- vs. distributed-memory, GPU, and their various permutations), efficient utilization, best practices
-  **CHAPEL**
 - fast, modern programming language for both shared- and distributed-memory systems
 - high-level, easy-to-use abstractions for data and task parallelism
 - one syntax across 3 levels of hardware parallelism: multi-core CPUs, multi-node clusters, and GPUs
 - hides the distinction between multi-threading and multi-processing from the programmer
 - PGAS (Partitioned Global Address space) programming model + computation follows data
 - focus on data parallelism through two numerical examples: one embarrassingly parallel (2D Julia set fractals) and one tightly coupled (heat diffusion) $\Rightarrow$ work on bottlenecks, load balancing, scaling
 - low-level interconnect communication hidden from the programmer (unlike in MPI)

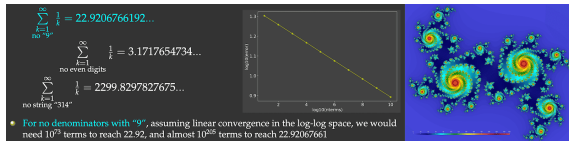
Parallel coding (continued)



- fast, high-level programming language for scientific computing and data science
- just-in-time compilation + interactive shell
- we take you on a journey to parallelize two problems: slow series and 2D Julia set
- multi-threading with Base.Threads and ThreadsX.jl
- multi-processing with Distributed.jl + arrays with DistributedArrays.jl and SharedArrays.jl
- work on thread safety, race conditions, benchmarking, load balancing



- aim: make it as fast + scalable as Julia or Chapel for numerical problems
- explore ~dozen different approaches: most fail for the slow series, and we analyze why
- eventually settle on a combination of Numba for compilation and Ray for distributed computing
- cover parallelizing with both Ray core and Ray data
- our latest GPU modules: Arrays with CuPy, Numba JIT compilation to CUDA kernels, DataFrames with Polars and RAPIDS



70 – 90 events each year across all formats

Current events at <https://training.westdri.ca> | Events | select the season

1. Biweekly webinars since March 2016

- alternating between advanced and introductory topics; **only topics not already covered in our courses**
- focus on new and/or impactful open-source tools and techniques that will benefit researchers
- most webinars delivered by the two of us; last year partnered with UBC
- ~ 150 past webinars archived at <https://training.westdri.ca> (chronologically and by topic)
- this term, webinars resumed in April to accommodate the Winter Visualization Series
<https://folio.vastcloud.org/winterseries>
- winter/spring 2026 training schedule
<https://training.westdri.ca/events/upcoming-training-winter-spring-2026>

2. Weekly online courses since February 2023

3. Annual week-long summer schools since 2017

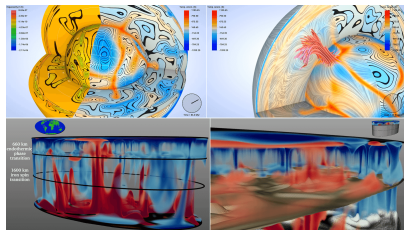
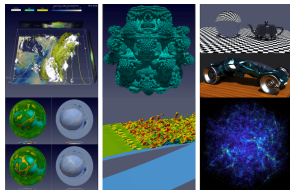
- typically every June @SFU: 5 full days, in-person at the Big Data Hub, ~ 70 registrants
- ideally, we would run a couple per year and distribute them geographically; previously organized and taught these at UBC, UVic, U of Manitoba, U of Saskatchewan, U of Calgary, UNBC
- this year June 1-5 @SFU (registration will open in April) © tell us what topics you'd like us to cover

4. National and international training events: HSS/Visualization Winter Series, national courses, DHISI (since 2016), multi-year PIMS/M2PI collaboration, IHPCSS

5. Local workshops and bootcamps available upon request for departments and research groups

Researcher support and outreach

- Official support support@tech.alliancecan.ca from the Alliance, mostly for specific problems on our clusters $\Rightarrow$ tickets answered by HPC / ARC analysts across the country
- Followup from our training, support emails and tickets
- Visualization support
 - I lead the National Visualization Team <https://ccvis.netlify.app>
 - answer most visualization tickets in the national queue
 - HPC background $\Rightarrow$ can visualize very large (GBs to TBs and larger) datasets
 - recent sci-vis projects with researchers at UBC (in-situ visualization), UQAM (climate model visualization), UVic (climate CFD, optimization), UofA (engineering CFD)
 - happy to help with any visualization projects
- Near-annual *Visualize This!* contests
 - nationally since 2016 <https://ccvis.netlify.app/contests>
 - in 2021 hosted the international IEEE Vis Contest <https://scivis2021.netlify.app>
 - currently on pause: we have data and technical expertise; seeking collaborative outreach partners
- Subscribe to our weekly mailing list <https://training.westdri.ca/contact>



Opportunities for you

- We offer training on any research computing topic in any format mentioned earlier
 - we constantly develop new materials, can look at new topics and specific computational problems
 - couple of restrictions
 1. in general, we do not cover commercial (not open-source) tools
 2. we probably do not have specialized knowledge in your field $\Rightarrow$ with few exceptions, cannot teach domain-specific topics & tools, but we could develop a joint course with you (domain knowledge from you and computational expertise from us)
- We can teach guest lectures or collaborate on graduate-level university courses
- Email training@westdri.ca to request a workshop or suggest a topic
- All our training is free for Canadian academic researchers

Other

- For other RCG offerings, check <https://docs.alliancecan.ca> | Services
 - HPC
 - cloud
 - database servers
 - Globus file transfer, Nextcloud storage
 - research software
- Special interest in numerical methods
 - ⇒ interested in collaboration on courses and materials
 - computational fluid dynamics, conservation methods, Riemann solvers
 - numerical radiative transfer (all regimes from free streaming to diffusion, 6D)
 - serial / parallel ODE and PDE solvers, stiff systems and implicit methods
 - adaptive techniques
 - ...